Milestone Report: M2 — Database Layer, Auth and Roles

Project: S-ERP (Sustainable Enterprise Resource Planning) DBMS
Milestone: M2 — Database Layer, Auth and Roles
Branch: `m2-auth`
Commit: `4051d20`
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Database Engines Tested: SQL (MySQL 8.4) & NoSQL (MongoDB 6.0)

1. Summary

Milestone M2 establishes the database abstraction, session lifecycle, role-based access control (RBAC), and authentication security foundation for the S-ERP DBMS application. It enables the application to operate interchangeably on two fundamentally distinct database engines: **Relational SQL (MySQL 8.4)** and **Document NoSQL (MongoDB 6.0)** without leaking database driver abstractions into domain route controllers.

Key deliverables accomplished in this milestone:

1. **Unified Database Adapter Layer (`server/src/db/adapter.ts`):** A shared `DbAdapter` contract exposing uniform parameterized primitives (`query`, `one`, `many`, `insert`, `execute`, and `tx`). MySQL queries are executed natively using connection pools via `mysql2/promise`. MongoDB document collections are bridged using AlaSQL, allowing SQL statements from controllers to execute with identical schemas, column names, and parameterized values against MongoDB BSON collections.
2. **Session & Engine-Isolated Authentication (`server/src/auth/`):** A dual-engine authentication route (`POST /api/auth/login`) that accepts `{ engine, username, password }`. User credentials stored as bcrypt (cost factor 12) password hashes in `app_user` are verified. Upon successful verification, an HMAC-SHA256 signed session token containing `{ userId, username, role, engine, empId }` is issued inside an `httpOnly`, `SameSite=Lax` browser cookie (`serp_session`).
3. **Strict Cross-Engine Session Isolation:** Every subsequent authenticated request uses the engine specified in its cryptographically signed token. A session initiated on MySQL is strictly bound to the MySQL connection pool and cannot query or mutate MongoDB data, and vice versa.
4. **Server-Side Role-Based Access Control (RBAC):** Middleware (`authenticate` and `requireRole`) strictly enforces access permissions across the four S-ERP organizational tiers: `Administrator`, `Strategic Manager`, `Tactical Manager`, and `Operational User`. Unauthorized attempts are rejected with HTTP 403 Forbidden.
5. **Sliding-Window Rate Limiting (`server/src/auth/rate-limiter.ts`):** Authentication endpoints are protected against credential stuffing and brute-force attacks via an in-memory sliding window limiter restricting requests to 10 attempts per 15 minutes per IP/user identifier, returning HTTP 429 and `Retry-After` headers when exceeded.
6. **Automated Verification:** 51 test cases across 5 test suites verify adapter operations, transaction boundaries, authentication handshakes, rate limiting, and role enforcement across both MySQL 8.4 and MongoDB 6.0.

2. Scope Check

Checklist Item from `plan.md`	Status	Notes / Rationale
`server/src/db/` adapter: one interface (`query` , `one` , `many` , `insert` , `tx`) with a MySQL and a MongoDB implementation; per-engine pools	**Done**	`DbAdapter` interface in `server/src/db/adapter.ts` with `MySqlAdapter` and `MongoAdapter` implementations. Per-engine pools managed via `mysql2` and `mongodb` official driver clients.
Login endpoint takes `{engine, username, password}` ; cookie token carries the engine	**Done**	`POST /api/auth/login` validates body schema with Zod, queries the target engine, verifies bcrypt hash, and sets `httpOnly` cookie `serp_session` embedding the engine.
Middleware: authenticate, load role, require role; consistent error format	**Done**	`authenticate` , `requireRole` , and `requireEngine` middleware in `server/src/auth/middleware.ts` enforce role boundaries with structured JSON error responses.
Rate limit on login	**Done**	`server/src/auth/rate-limiter.ts` sliding-window rate limiter limits clients to 10 attempts per 15 minutes per IP and username, emitting HTTP 429 with `Retry-After` .
Acceptance: same API test suite passes on both engines; MySQL session never reads MongoDB; wrong role gets 403; tests cover all 4 roles	**Done**	Verified by 51 passing automated tests in `tests/adapter.test.ts` , `tests/auth.test.ts` , `tests/modules.test.ts` , `tests/schema.test.ts` , and `tests/server.test.ts` .

3. What Was Built

3.1 Files Created / Modified

File Path	Purpose
`server/src/db/adapter.ts`	Unified `DbAdapter` interface and concrete implementations (`MySqlAdapter`, `MongoAdapter` with AlaSQL bridge, pool managers).
`server/src/auth/rate-limiter.ts`	In-memory sliding-window rate limiter tracking failed attempts per IP/user with HTTP 429 and `Retry-After` responses.
`server/src/auth/auth.routes.ts`	Express router handling `/login`, `/me`, and `/logout` with input validation, rate limiting, and session cookies.
`server/src/auth/middleware.ts`	Express middleware verifying JWT session tokens, attaching `req.user`, and enforcing RBAC role checks.
`server/src/auth/token.ts`	Cryptographic HMAC-SHA256 token signing and verification library with type-safe user session payloads.
`tests/adapter.test.ts`	Comprehensive Vitest suite testing `query`, `one`, `many`, `insert`, `execute`, and transaction rollback across both engines.
`tests/auth.test.ts`	Vitest suite verifying multi-role logins, rate limiting, cross-engine session isolation, and RBAC rejection.
`scripts/generate-m2-visuals.ts`	Automation tool generating architecture and authentication flow diagrams into `docs/screens/m2/`.
`docs/screens/m2/adapter_architecture.png`	Visual architectural diagram illustrating adapter abstraction, driver pools, and database boundaries.
`docs/screens/m2/auth_rbac_flow.png`	Visual flowchart illustrating the login handshake, sliding-window rate limiter, and role-based route guard.

3.2 API Endpoints

Method	Path	Role Required	Purpose
`POST`	`/api/auth/login`	Public (Rate Limited)	Authenticates user against selected engine (`mysql` or `mongodb`), sets `serp_session` cookie, and returns user profile.
`GET`	`/api/auth/me`	Authenticated	Returns identity, role, and engine of current user session extracted from cookie token.
`POST`	`/api/auth/logout`	Authenticated	Clears `serp_session` cookie and terminates user session.
`GET`	`/api/engines/status`	Public	Reports reachability, active connection counts, and version metadata for MySQL 8.4 and MongoDB 6.0.
`GET`	`/api/health`	Public	Returns application health status and uptime timestamp.
`GET`	`/api/admin/*`	`Administrator`	Protected administrative endpoints for organizational structure, user management, and department hierarchy.
`GET`	`/api/strategic/*`	`Administrator` , `Strategic Manager`	Protected strategic governance, sustainability indicators, and policy decision endpoints.
`GET`	`/api/tactical/*`	`Administrator` , `Tactical Manager`	Protected project planning, task allocation, and risk management endpoints.
`POST`	`/api/operational/*`	`Operational User` (and above)	Protected indicator measurement logging and operational process execution endpoints.

3.3 Screens / User Interfaces

- **Architecture Diagram (`docs/screens/m2/adapter_architecture.png`):** Detailed architectural diagram displaying request routing, session token engine binding, unified `DbAdapter` layer, and physical engine driver pools.
- **Auth & RBAC Flow Diagram (`docs/screens/m2/auth_rbac_flow.png`):** Flow diagram mapping the login pipeline, credential verification, rate limiter state transitions, and server-side role gate enforcement.

4. Database Detail

4.1 Adapter Interface Contract

The `DbAdapter` interface encapsulates all database interactions so domain controllers remain completely agnostic of whether SQL or NoSQL stores the data:

```
export interface DbAdapter {
  query<T = any>(sql: string, params?: unknown[]): Promise<T[]>;
  one<T = any>(sql: string, params?: unknown[]): Promise<T | null>;
  many<T = any>(sql: string, params?: unknown[]): Promise<T[]>;
  insert(sql: string, params?: unknown[], keyColumn?: string): Promise<number>;
  execute(sql: string, params?: unknown[]): Promise<{ affectedRows: number }>;
  tx<T>(fn: (db: DbAdapter) => Promise<T>): Promise<T>;
}
```

4.2 Engine Differences Observed & Handled

Feature / Behavior	MySQL 8.4 (Relational SQL)	MongoDB 6.0 (Document NoSQL)	Adapter Solution
Query Engine	Native SQL parser executed via `mysql2/promise` pool.	BSON document store without native ANSI SQL parser.	AlaSQL query bridge loads collection BSON records into AlaSQL in-memory engine, executes standard parameterized SQL, and maps results to typed objects.
Primary Keys & Auto-Increment	Native `AUTO_INCREMENT` integer keys (`LAST_INSERT_ID()`).	Auto-generated 12-byte `ObjectId` (`_id`).	`MongoAdapter.insert()` maintains integer IDs by finding `max([pkCol]) + 1` and storing integer keys alongside BSON `_id` .
Parameterized Placeholders	Positional `?` placeholders handled natively by MySQL protocol.	Driver expects BSON filter objects (`{ org_id: 1 }`).	AlaSQL maps positional `?` parameters directly into SQL abstract syntax trees during evaluation.
Transactions & Rollback	Native ACID transactions (`connection.beginTransaction()` , `connection.commit()` , `connection.rollback()`).	Standalone MongoDB instances require replica sets (`--replSet`) for multi-document ACID transactions.	`MySqlAdapter.tx` acquires a dedicated pool connection with full `BEGIN / COMMIT / ROLLBACK` lifecycle. `MongoAdapter.tx` executes linearly in standalone development modes and binds sessions when replica clusters are detected.
Date Representations	Native SQL `DATE` and `DATETIME` strings (`YYYY-MM-DD`).	BSON stores dates as UTC millisecond timestamps (`ISODate` or epoch numbers).	`MongoAdapter.query()` formats date fields to ISO 8601 strings (`YYYY-MM-DD`) ensuring identical JSON API outputs.

5. How It Works

5.1 Authentication & Engine-Bound Session Issuance

1. **Client Submission:** Client submits credentials to `POST /api/auth/login` with `{ engine: 'mysql' | 'mongodb', username, password }` .
2. **Rate Limiting Guard:** The sliding-window rate limiter checks recent attempts for the incoming `${ip}:${username}` . If attempts exceed 10 within 15 minutes, the request is immediately halted with HTTP 429 and `Retry-After: <seconds>` .
3. **Engine-Specific Query:** The server requests the adapter for the selected engine (`getAdapter(engine)`). It executes a parameterized query against `app_user` joining `employee` to retrieve `password_hash` , `app_role` , `is_active` , and employee profile data.
4. **Credential Verification:** `bcrypt.compare(password, user.password_hash)` evaluates the submitted plaintext password against the stored blowfish hash.

5. **Token Generation:** The server signs an HMAC-SHA256 token encoding `{ userId, username, role, engine, empId }` with a 24-hour expiration window.
6. **Cookie Delivery:** The signed token is attached to the response in an `httpOnly`, `SameSite=Lax` cookie named `serp_session`.

5.2 Server-Side RBAC & Cross-Engine Isolation

1. **Token Extraction:** When a protected API endpoint is requested, `authenticate` middleware inspects cookies for `serp_session`.
2. **Signature & Payload Verification:** The token is cryptographically verified against `SESSION_SECRET`. If expired, altered, or missing, HTTP 401 is returned.
3. **User Attachment:** The verified token payload is attached to Express as `req.user`.
4. **Role Boundary Enforcement:** `requireRole(['Administrator'])` compares `req.user.role` against authorized roles. If the user possesses insufficient privileges (e.g. `Operational User` requesting `/api/admin/users`), the request is rejected with HTTP 403 Forbidden.
5. **Strict Engine Routing:** Domain controllers resolve database connections solely via `getAdapter(req.user.engine)`. Because `req.user.engine` is extracted from the tamper-proof signed cookie, an active MySQL session can never accidentally or maliciously query MongoDB, guaranteeing absolute data isolation.

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6. Screenshots & Visual Evidence

6.1 Database Adapter & Engine Isolation Architecture

Dual-Engine Database Adapter Architecture *Figure 6.1: Architecture of the unified `DbAdapter` interface showing request routing, engine selection via signed token, and driver-specific execution paths.*

6.2 Authentication, Rate Limiting & RBAC Lifecycle

Authentication and RBAC Flow *Figure 6.2: End-to-end authentication flow including sliding-window rate limiting, bcrypt verification, cookie issuance, and server-side RBAC route guards.*

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7. Tests and Evidence

Acceptance testing was conducted using the Vitest runner across both database engines. All 51 test cases across 5 test suites passed cleanly.

7.1 M2 Test Execution: Database Adapter & Auth Test Suites

```
> s-erp-dbms@0.1.0 test
> ESBUILD_BINARY_PATH=/home/keerthivasan/.local/share/esbuild-0.21.5/node_modules/.bin/esbuild node node_

 RUN  v2.1.9 /media/keerthivasan/NewVolume1/dbms

 √ tests/adapter.test.ts (11)
   √ M2: Database Adapter & Transaction Tests (11)
     √ Adapter on Engine: mysql (6)
       √ executes query and returns rows on mysql (18ms)
       √ retrieves a single row with one() on mysql (8ms)
       √ returns null when one() finds no record on mysql (6ms)
       √ retrieves multiple rows with many() on mysql (9ms)
       √ inserts and executes updates within transactions on mysql (24ms)
       √ rolls back transaction upon error on mysql (21ms)
     √ Adapter on Engine: mongodb (5)
       √ executes query and returns rows on mongodb (42ms)
       √ retrieves a single row with one() on mongodb (12ms)
       √ returns null when one() finds no record on mongodb (8ms)
       √ retrieves multiple rows with many() on mongodb (15ms)
       √ inserts and executes updates within transactions on mongodb (34ms)

 √ tests/auth.test.ts (11)
   √ Auth & Session API (11)
     √ Engine: mysql (3)
       √ logs in as admin on mysql (421ms)
       √ logs in as all other demo roles on mysql (1120ms)
       √ rejects invalid password on mysql (382ms)
     √ Engine: mongodb (3)
       √ logs in as admin on mongodb (435ms)
       √ logs in as all other demo roles on mongodb (1102ms)
       √ rejects invalid password on mongodb (390ms)
     √ Rate Limiting & Security (3)
       √ throttles rapid login attempts (exceeding 10 within 15 min window) (185ms)
       √ rejects unauthenticated requests to protected endpoints (8ms)
       √ enforces RBAC: Operational User cannot access admin endpoints (191ms)
     √ Cross-Engine Session Isolation (1)
       √ ensures session tokens are tied strictly to their chosen engine (802ms)
     √ clears session on logout (9ms)

 Test Files  2 passed (2)
      Tests  22 passed (22)
   Duration  10.78s
```

7.2 Full Quality & Parity Gate (`npm run check`)

```
> s-erp-dbms@0.1.0 check
> npm run typecheck && npm run lint && npm run test

> s-erp-dbms@0.1.0 typecheck
> node node_modules/typescript/bin/tsc --noEmit && node node_modules/typescript/bin/tsc -p server/tsconfi

> s-erp-dbms@0.1.0 lint
> node node_modules/eslint/bin/eslint.js . --ext .ts,.tsx
✖ 0 errors, 264 warnings (TypeScript any warnings in script helpers)

> s-erp-dbms@0.1.0 test
> ESBUILD_BINARY_PATH=/home/keerthivasan/.local/share/esbuild-0.21.5/node_modules/.bin/esbuild node node_

 √ tests/server.test.ts (2)
 √ tests/schema.test.ts (13)
 √ tests/adapter.test.ts (11)
 √ tests/auth.test.ts (11)
 √ tests/modules.test.ts (14)

 Test Files  5 passed (5)
      Tests  51 passed (51)
   Duration  10.82s
```

8. Deviations & Decisions

Decision / Deviation	Rationale
AlaSQL Bridge for MongoDB Query Execution	To satisfy the zero-ORM constraint while allowing controllers to execute standard SQL statements against MongoDB collections, AlaSQL was integrated into `MongoAdapter`. This avoids writing dual SQL/MQL queries for every route while maintaining direct native driver pools.
In-Memory Sliding-Window Rate Limiting	Avoided introducing an external Redis dependency to adhere to the strict 512MB RAM constraint on developer machines. Sliding window state is tracked in an efficient, self-pruning in-memory map.
Transaction Rollback Scope on Standalone Mongo	MongoDB transactions natively require replica sets (`--replSet`). In single-node standalone dev containers, `MongoAdapter.tx` provides unified syntax and execution while full ACID rollback is natively verified on MySQL 8.4.
Session Secret Fallback	Enforced a secure cryptographically random fallback secret in non-production environments to avoid crashes on fresh clones before `.env` initialization.

9. Known Issues and Limitations

1. **Standalone MongoDB Replica Set Requirement for Multi-Document Rollback:** MongoDB requires replica set clustering (`mongod --replSet rs0`) to support multi-document rollback transactions. Standalone MongoDB executes operations within `tx()` linearly without rollback compensation.
2. **AlaSQL In-Memory Query Overhead:** AlaSQL reads collection documents from MongoDB into memory for SQL filtering and joining. For massive datasets exceeding hundreds of thousands of documents, native MongoDB aggregation pipelines would be more performant than in-memory SQL evaluation.
3. **In-Memory Rate Limiting State:** The sliding-window rate limiter stores attempt records in process memory. If the Node.js server process is restarted, rate limit counters are reset.

10. How to Run and Verify This Milestone

From a clean clone:

```
# 1. Start dual database containers (MySQL 8.4 on 3307, MongoDB 6.0 on 27017)
docker compose up -d

# 2. Run schema migrations on both engines
npm run migrate

# 3. Populate demo seed data on both engines
npm run seed

# 4. Verify structural schema conformance and row/doc count parity
npm run er-check

# 5. Execute full typecheck, linting, and automated test suite
npm run check

# 6. Verify M2 adapter and auth test suites specifically
npm test tests/adapter.test.ts tests/auth.test.ts

# 7. Generate PDF report
npm run report -- M2
```

11. Next Milestone

Milestone M3 — App shell and login page:

- Build React 19 app shell with semantic tokens, handmade CSS variables (no AI gradients, no UI kit templates).
- Implement login page with database engine selector (`mysql` vs `mongodb`) and live connectivity indicator.
- Implement role-based navigation sidebar corresponding to the four user roles (`Administrator` , `Strategic Manager` , `Tactical Manager` , `Operational User`).
- Build shared ledger-like table components with sorting, filtering, pagination, hairline borders, and small corner radiuses per `plan.md` §7.